

BobaLink Browser Extension “ Status

Revision: 4 **Last modified:** 2026-06-10T20:40:00Z **Scope:** BobaLink (extension/) “ WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs and forwards them to the Boba merge service on port 7187. **Authority:** master plan docs/browser_extension/IMPLEMENTATION_PLAN.md (9 phases).

Captured-evidence-driven (11.4.5 / 11.4.45). Every PASS cites a real commit hash and/or a verified test/file artifact. Rows lacking runtime evidence are marked IN-PROGRESS or PENDING “ never PASS (11.4.6 no-guessing).

Baseline facts (verified this session “ Session 11)

- **HEAD:** 2011810 (pushed); wave-4 (Phase-9 ci-ext.sh gate + per-store zips, ru locale, content-XSS, credential-leak audit, decrypt-throw parity, tab-group Challenge, and the _locales packaging fix) staged for the next commit.
- **Extension test corpus:** 413 Vitest tests across 37 spec files “ same-session npx vitest run “ Tests 413 passed (413) “ under extension/tests/{unit,perf,stress,chaos,integration,security,e2e} + src/**/*.npx tsc --noEmit clean; npm run lint 0 errors / 0 warnings. (+126 tests over the prior 287/22 baseline: integration 7, security 4 files, stress/chaos 10, locale 4, token suites 8, tab-groups 13, ally 18, group-send nits 2, +more.)
- **WXT build wiring “ COMPLETE.** WXT entrypoints exist at extension/src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html} (thin wrappers over the existing logic modules, so all prior test imports stay valid). npx wxt build produces a **loadable** extension/.output/chrome-mv3/ AND firefox-mv2/:manifest_version:3, background.service_worker:background.js, popup.html, options.html, content-scripts/content.js, icons 16/32/48/128, AND _locales/{en,ru}/messages.json (REQUIRED because default_locale:"en" + __MSG_*__ “ see the 11.4.138 note below). **Every manifest-referenced asset + the default-locale catalog verified present on disk** by the ci-ext.sh manual gate (11.4.38). Content-script matches are derived from the curated SITE_SELECTORS (24 hosts, **no** <all_urls>); permissions least-privilege (no scripting/tabs/cookies; +tabGroups for Phase-5); host_permissions = http://localhost:7187/* only; CSP script-src 'self'. Per-store zips (boba-link-1.0.0-{chrome,firefox,sources}.zip) produced by ci-ext.sh.

Per-phase status

Phase	Title	Status	Evidence
1	Foundation & scaffolding (WXT config, TS, shared libs, types, constants)	PASS	Scaffolded @33a9815; shared-lib + constants + type-guard unit specs green @15a9a61
2	Core detection & parsing engine (parsers, scanners, infohash dedup)	PASS	Parsers @7225470/fa03323; SHA-1 infohash + link/text scanners + perf/stress @fa03323;

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			scanner orchestrator cross-scanner dedup @946c61e
3	Extension shell “content / background / popup / options	PASS	Shell @e8fde43; background service-worker message router capstone @15a9a61
â€”	WXT build wiring (entrypoints/, wxt build “ .output/, manifest validate)	PASS	Entrypoints src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html}; npx wxt build “loadable .output/chrome-mv3/ (8/8 manifest assets verified present, Â§11.4.38); matches derived from SITE_SELECTORS, no <all_urls> (Session 11 @5edf6ac, pushed)
4	Boba backend integration (real client, queue auth, BE-1/2/3)	IN-PROGRESS	API leaf @e8fde43. Phase-7 decrypt-before-send wired (Session 11): BobaClient.create() decrypts the encryptedBobaApiToken bundle via shared/crypto; background reads the session passphrase from chrome.storage.session and sends the decrypted plaintext (default-open when locked). Token suites boba-client-token.test.ts 5 + background-token.test.ts 3 green. PENDING: live-7187 integration (require_backend(7187)), end-to-end detect“send“torrent-in-qBittorrent on the real backend
			Wired (Session 11): src/tabgroups/index.ts (dedupe across a group + batch dispatch, 13 tests) integrated into background/index.ts MENU_SEND_GROUP (deduped group batch “one addMagnets POST); manifest +tabGroups (MINIMAL “research-confirmed tabs NOT needed since only tab.id is read; Â§11.4.120 security-test reconciliation, mutation-verified). New background.test.ts MENU_SEND_GROUP test (RED-verified: handler no-op “FAIL). Independent review: GO-with-nits . Group-send nits (a) offline-queue enqueue-on-failure, (b) network-error notification, (c) hardened async flush “all FIXED (Session 11, RED“GREEN + 3—deterministic, independent review GO). TRACKED: decrypt-throw (wrong passphrase) on group send not enqueued/notified “pre-existing parity with Send-All, future phase
5	Tab-group batch (chrome.tabGroups “per-tab scan “batch send)	IN-PROGRESS	il8n: locale.test.ts (en completeness) + ru locale added with locale-parity.test.ts (en“ru key + placeholder parity, mutation-proven); ally: tests/ally/
6	UI/UX, il8n, accessibility, themes	IN-PROGRESS	

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7 Security & credentials (delegate-by-default, no embedded key, log redaction)	IN-PROGRESS	<code>{popup,options}.ally.test.ts</code> (18 tests â€” roles/accessible-names/tablistâ€”i,Žtabpanel/live-regions, mutation-proven). PENDING: 6 more locales (plan targets 8), deeper WCAG (contrast/focus/keyboard), theme-switch evidence <code>crypto.ts</code> adopted; decrypt-and-send path landed + decrypt-throw on wrong passphrase now enqueues+notifies (parity, REDâ†’GREEN); session passphrase from <code>chrome.storage.session</code>, no embedded key, plaintext/passphrase never logged; security suite <code>tests/security/*</code> (least-privilege manifest, CSP, no-hardcoded-secret, secret-storage, content-XSS â€” render path proven safe via <code>innerHTML-mutation</code>â†’FAIL); Â§11.4.10.A credential-leak audit challenges/<code>security/credential_leak_audit.sh</code> (PASS, mutation-verified). PENDING: full pen-test suite (sender-origin validation, rate-limit) 413 tests/37 files green (unit/perf/stress/chaos/integration/security/e2e). Challenge <code>challenges/extension/detect_and_forward_challenge.sh</code> drives the REAL orchestrator+client end-to-end, PASS on captured evidence, mutation-verified (no-op stub â†’ FAIL). HelixQA boba-bobalink.yaml BOBA-LINK-007 (detectâ†’forward) + BOBA-LINK-008 (tab-group batch) added; symlinked into <code>challenges/helixqa-banks/.tab-group+detectâ†’forward</code> Challenges both mutation-verified. E2E <code>tests/e2e/extension-loads.spec.ts</code> is a real MV3-load test, operator-gated SKIP in this headless sandbox (extension load unsupported; Â§11.4.3). PENDING: the full 13-type coverage matrix + live-backend integration + the coverage ledger to 100% Manual gate <code>extension/ci-ext.sh</code> (Session 11, Â§11.4.18 doc'd): <code>tsc</code> â†’ lint â†’ full vitest â†’ chrome+firefox builds â†’ Â§11.4.38 artifact-verify (opens the manifest, asserts every referenced asset + the <code>default_locale</code> catalog exist non-zero) â†’ per-store <code>wxt zip</code>. CI-EXT: PASS â€” produces loadable <code>chrome-mv3/ + firefox-mv2/ + bobalink-1.0.0-{chrome,firefox,sources}.zip</code>. PENDING: store-listing metadata/submission + Â§11.4.65 user/dev/install doc siblings. NO CI/CD (manual only)
8 Testing to 100% (all types) + Challenges + HelixQA	IN-PROGRESS	
9 Build, packaging & distribution (manual â€” NO CI/CD)	IN-PROGRESS	

Status legend

- **PASS** “ implemented and backed by a cited commit and/or verified test/file evidence.
- **IN-PROGRESS** “ partially landed; some sub-tasks done with evidence, remainder explicitly enumerated as PENDING.
- **PENDING** “ not yet started / no runtime evidence; planned in `IMPLEMENTATION_PLAN.md`.

Anti-bluff notes (Â§11.4 / Â§11.4.6 / Â§11.4.69)

- The 413 figure IS a same-session recorded `npx vitest run` result (`Tests 413 passed (413)`), not merely a static `grep` “ it supersedes the prior 287/22 `grep`-count baseline. `tsc + lint` captured clean in the same session.
- The Â§11.4.38 loadable-artifact claim is verified by `ci-ext.sh` **opening the produced** `.output/chrome-mv3/manifest.json` and confirming every referenced asset AND the `default_locale` catalog exist non-zero on disk “ not by “build exited 0”.
- **Â§11.4.138 bluff-audit (corrected this session)**. An earlier Status revision claimed the artifact was loadable on an “8/8 referenced assets present” check. That check was INCOMPLETE: the manifest declares `default_locale: "en" + __MSG_*__`, but the build packaged **no** `_locales/` (WXT copies static assets from `src/public/`, while the catalog lived at `src/_locales/`) “ so Chrome would have REJECTED the extension at load. Caught by the new `ci-ext.sh` Â§11.4.38 step (which checks the `default_locale` “ `_locales/` invariant). Root cause fixed by moving the catalog to `src/public/_locales/{en,ru}/`; the rebuilt artifact now packages it (verified on disk). Permanent guard = `ci-ext.sh` (the regression can't recur silently). This is the Â§11.4.108 lesson: `build-exit-0` “ deployable-artifact.
- The Phase-7 decrypt path is proven by RED“GREEN: ciphertext-on-the-wire (pre-fix) “ decrypted-plaintext header (post-fix), with the ciphertext never sent and the token never logged (`background-token.test.ts`, `boba-client-token.test.ts`).
- The detect“forward Challenge was independently re-run and its no-op-stub negation verified (stubbed orchestrator “ FAIL) during code review.
- Independent multi-lens subagent code-review (correctness/security/build/docs) was run; the security + correctness + build + docs lenses returned GO / GO-with-nits. (The correctness/security/build subagent lenses were re-run in the conductor context after a transient platform throttle on subagent dispatch; an independent subagent re-pass runs before any release tag per Â§11.4.40.)